

# Cerulopam — Commercial Stability Protocol

Protocol STAB-P-2021-61

## Data Evaluation and Reporting

Stability data are evaluated per ICH Q1E. Regression analysis is applied to quantitative attributes exhibiting change over time; poolability across batches is assessed at the 0.25 significance level. Annual reports summarize all data generated in the reporting interval.

## Study Storage Conditions

The primary (long-term ICH condition) storage condition for this protocol is  $25^{\circ}\text{C} \pm 2^{\circ}\text{C}$  /  $60\% \text{ RH} \pm 5\% \text{ RH}$ . Accelerated studies at  $40^{\circ}\text{C} \pm 2^{\circ}\text{C}$  /  $75\% \text{ RH} \pm 5\% \text{ RH}$  were completed through six months during development and are not part of this protocol.

## Testing Schedule

| Interval (months) | 0 | 3 | 6 | 9 | 12 | 18 | 24 |
|-------------------|---|---|---|---|----|----|----|
| Long-term testing | X | X | X | X | X  | X  | X  |

## Bracketing and Matrixing

A bracketing design per ICH Q1D tests the extremes of the strength range (20 mg and 100 mg); the 25 mg strength shares identical formulation composition and packaging and is therefore not tested.

## Stability Commitments

As a condition of approval, three consecutive production batches of Cerulopam will be entered into the stability program, and the retest period will be re-evaluated after 36-month data are available.

## Acceptance Criteria

The stability acceptance criterion applied at each interval is — Total degradation products: not more than 2.0%. All other registered specification tests apply unchanged.

## Deviations

The 9-month samples for batch 002 were held at ambient conditions for 11 hours during chamber maintenance before testing; trending showed no anomalous results.

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## **Batch Selection**

The protocol covers three primary batches manufactured at commercial scale using the approved process. Batch genealogy, manufacturing dates, and packaging lots are recorded in the batch appendix. All batches were released against the approved specification prior to set-down.